MASTER CLINICAL SUPPLY QUALITY AGREEMENT (Q-AGR)

Document Ref: Vertex Master Cell Therapy Transport Protocol (Q-AGR-VTX-88) | **Standard:** FDA 21 CFR Part 11 & EU GMP Annex 15

1. Commercial Parties & Asset Specifications

Commercial Sponsor	Vertex Pharmaceuticals
Sponsor Identifier	CUST-VTX-02
Therapeutic Asset	**Casgevy (exagamglogene autotemcel) CRISPR Cell Therapy**
Active Batch / Lot #	LOT-CAS-8812-B
Commercial Valuation	**$580,000 USD**
Consignee Medical Facility	Memorial Sloan Kettering Cancer Center

2. Storage Specifications & Excursion Limits

The carrier is legally and operationally bound to preserve custody within qualified physical boundaries under Tier **Ultra-Low Cryogenic Liquid Nitrogen SLA (<-150°C) (SLA-CRYO-LN2)**. Continuous edge telemetry samples temperature at ≤ 1.0 second frequency with cryptographically verifiable records.

Qualified Temperature Band	**-196.0°C to -150.0°C**
Critical Regulatory Ceiling	-150.0°C
Critical Regulatory Floor	2.0°C
Allowable Cumulative Excursion	30 minutes maximum
Sampling & Seal Frequency	Real-Time Telematics (1.0 sample/s)

3. Financial Liability & Liquidated Damages Schedule

Failure to remediate an active physical breach beyond -150.0°C within 30 minutes constitutes complete batch degradation, triggering full commercial reimbursement of **$580,000 USD**. For delays exceeding contract cutoff without thermal breach, liquidated damages accrue at **$75,000 operating theater cancellation penalty**.

4. Autonomous Triage & Emergency Diversion Protocols

Under FDA 21 CFR Part 11 electronic control provisions, the commercial sponsor confers autonomous executive authority to **CryoGuard AI™**. In the event of primary equipment failure, the system is authorized to:

- Perform multi-criteria facility scoring (distance, bay capacity, certified capabilities).
- Execute closed-loop highway diversion to the nearest compatible secondary biorepository.
- Automatically generate cryptographically signed CAPA non-conformance reports.

5. Regulatory Certification & Electronic Signatures

For Commercial Sponsor: Vertex Pharmaceuticals *Digitally Signed: e-Sig SHA-256* Date: 2026-10-14	**For Logistics Carrier & AI Operator:** CryoGuard Autonomous Logistics Inc. *Automated Execution Authority Active* Date: 2026-10-14